

1. What is Statistical Distribution?

A statistical distribution is a mathematical function that describes how data values are spread across different ranges. It helps understand the pattern, frequency, and probability of data occurrences in a dataset. Statistical distributions are widely used in data analysis, probability theory, machine learning, economics, healthcare, and e-commerce systems. By studying distributions, analysts can identify whether data is normally distributed, skewed, concentrated, or spread out.

In this project, transaction amounts are analysed using statistical distributions to understand customer purchasing behaviour. Some customers make small purchases while a few customers make very large purchases, creating a skewed distribution.

Formula

Probability Distribution Function:

$$P(X = x)$$

Where:

- $P(X = x)$ = Probability of random variable X
- x = Data value

Real-World Example

In an e-commerce platform, most customers purchase products worth ₹500–₹2000, while only a few customers make purchases above ₹10000. This creates a positively skewed transaction distribution.

2. What is a Q-Q Plot and Why is it Used?

A Q-Q Plot (Quantile-Quantile Plot) is a graphical method used to compare the distribution of a dataset with a theoretical normal distribution. It helps determine whether data follows a normal distribution. In a Q-Q plot, if the plotted points lie close to the straight diagonal line, the data is approximately normal. If points deviate from the line, the data is skewed or contains outliers.

In this project, Q-Q plots help analyse whether transaction amounts follow normal distribution before applying transformations.

Formula

Quantile Comparison:

$$Q_{sample} = Q_{normal}$$

Real-World Example

An e-commerce company checks whether customer spending patterns are normally distributed before building a financial forecasting model.

3. Difference Between Discrete and Continuous Distribution

Discrete distributions represent countable values such as number of transactions, number of customers, or failed payments. Continuous distributions represent measurable values such as transaction amounts, delivery times, or customer income.

Discrete distributions have separate individual values, while continuous distributions contain values within an interval.

Formula

Discrete Probability:

$$P(X = x)$$

Continuous Probability:

$$P(a < X < b)$$

Real-World Example

- Discrete: Number of successful transactions per day
 - Continuous: Transaction amount paid by customers
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4. What is Bernoulli Distribution?

Bernoulli Distribution is a probability distribution with only two possible outcomes:

- Success
- Failure

It is commonly used in binary classification problems and yes/no situations. In this project, transaction status (Success/Fail) follows Bernoulli distribution.

Formula

$$P(X = x) = p^x(1 - p)^{1-x}$$

Where:

- p = Probability of success
- $1 - p$ = Probability of failure

Real-World Example

An online payment either succeeds or fails during checkout.

5. What is Binomial Distribution?

Binomial Distribution measures the probability of obtaining a certain number of successful outcomes in a fixed number of independent trials. It extends Bernoulli distribution for multiple trials.

In this project, it can model the number of successful transactions out of 10 transactions.

Formula

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Where:

- n = Total trials
- x = Number of successes
- p = Probability of success

Real-World Example

Probability of 8 successful payments out of 10 online transactions.

6. Explain Log-Normal Distribution

A Log-Normal Distribution occurs when the logarithm of the data follows a normal distribution. It is commonly used for highly skewed data such as income, stock prices, and transaction amounts.

In this project, transaction amounts follow log-normal distribution because most purchases are small while a few are very large.

$$Y = \ln(X)$$

Real-World Example

Most customers buy products worth ₹1000–₹3000, but a few customers purchase premium items worth ₹50000.

7. Explain Power Law Distribution

Power Law Distribution describes situations where a small number of observations are extremely large compared to the rest. It creates a long-tail pattern.

In e-commerce systems, a few customers contribute very high transaction amounts while most customers contribute smaller amounts.

Formula

$$P(x) = Cx^{-\alpha}$$

Where:

- C = Constant
- α = Exponent

Real-World Example

A small number of online customers generate most of the company revenue.

8. What is Box-Cox Transform?

Box-Cox Transformation is a statistical technique used to reduce skewness and stabilize variance in data. It helps convert non-normal data into approximately normal data.

In this project, Box-Cox transformation improves transaction amount distribution for better statistical analysis.

Formula

$$Y(\lambda) = \frac{X^\lambda - 1}{\lambda}$$

Real-World Example

An e-commerce analyst transforms highly skewed sales data before building predictive models.

9. Explain Poisson Distribution with Example

Poisson Distribution models the probability of events occurring within a fixed interval of time or space. It is used for count-based data.

In this project, it models the number of transactions occurring per day.

Formula

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Where:

- λ = Average event rate

Real-World Example

An e-commerce platform receives an average of 20 transactions per hour.

10. What is Z-score Probability?

Z-score measures how far a data value is from the mean in terms of standard deviations. It helps identify outliers and unusual values.

In this project, Z-score identifies unusually high transaction amounts.

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X = Data value
- μ = Mean
- σ = Standard deviation

Real-World Example

A transaction of ₹50000 may have a very high Z-score and could indicate unusual customer activity.

11. Difference Between PDF and CDF

PDF (Probability Density Function) shows the likelihood of data occurring at a specific value, while CDF (Cumulative Distribution Function) shows cumulative probability up to a certain value.

PDF explains density, while CDF explains accumulation.

PDF Formula

$$PDF = f(x)$$

CDF Formula

$$CDF = P(X \leq x)$$

Real-World Example

- PDF: Probability density of transactions around ₹3000
- CDF: Probability that transaction amount is less than or equal to ₹3000

Final Conclusion

The Statistical Distribution and Probability Analysis for E-commerce Transactions project successfully analysed customer transaction behaviour using probability distributions, statistical techniques, and data visualization methods.

The analysis showed that transaction amounts are positively skewed, meaning most customers make smaller purchases while a few customers make very large transactions. Because of this behavior, the Log-Normal Distribution and Power Law Distribution were found to best represent the transaction amount data.

Bernoulli and Binomial distributions were used to analyse transaction success and failure patterns. The results showed that most transactions were successful, indicating stable transaction performance within the e-commerce system.

Poisson Distribution effectively modelled transaction frequency and helped identify the average number of customer transactions occurring within a fixed period.

The Q-Q Plot demonstrated that the original transaction data did not follow a perfect normal distribution due to skewness and outliers. To improve data normality and reduce skewness, Box-Cox Transformation was applied successfully.

Z-score analysis helped identify unusual and high-value transactions by measuring how far transaction amounts deviated from the average transaction value.

PDF and CDF plots provided a clear understanding of transaction probability density and cumulative transaction behaviour across different transaction ranges.

The project demonstrates how statistical distributions and probability analysis can be used in real-world e-commerce systems to:

- Understand customer purchasing behaviour

- Detect abnormal transactions
- Analyse transaction success patterns
- Support business decision-making
- Improve financial and customer analytics

Overall, the project successfully applied probability theory, statistical distributions, and data visualization techniques to extract meaningful insights from e-commerce transaction data.